Foveated Diffusion Through a Neuroscience Lens: Reproducing Mixed-Resolution Generation on a Budget

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1 The original work: Foveated Diffusion

Modern text-to-image and video generators are Diffusion Transformers (DiTs): an image is encoded by a VAE into a latent grid, the latent is cut into $h \times w$ patch tokens, and a transformer iteratively denoises that token sequence. Because attention is quadratic in the number of tokens, higher resolutions, frame rates and context lengths quickly become prohibitively expensive. Chao *et al.* observe that the consumer of a generated image is a human, whose visual acuity is high only in a small region around the point of gaze (the fovea) and falls off rapidly with eccentricity. Foveated rendering in computer graphics has used this for over a decade; the paper brings the same idea to generative models.

Foveated tokenization. Given a binary mask $M \in \{0,1\}^{h\times w}$ on the token grid, tokens inside the fovea are kept at full resolution, and every 2×2 block of tokens outside it is replaced by a single low-resolution token (d=2, a 4× saving in the periphery). With m foveal tokens the sequence length becomes

$$L = m + \frac{h \cdot w - m}{4} \ll h \cdot w, \tag{1}$$

and the ratio $L/(hw)$ is reported as the *token ratio*. The DiT denoises this packed, variable-length sequence end to end; the full-resolution grid is never materialised.

Generation and decoding. Generation starts from Gaussian noise of length L, which the model iteratively denoises into a clean foveated sequence $z_0^{\text{fov}} \in \mathbb{R}^{c\times L}$. It is split into its high- and low-resolution parts, and each part is decoded separately by the VAE decoder $\mathcal{D}$:

$$(z_0^{\text{high}}, z_0^{\text{low}}) = \text{Split}(z_0^{\text{fov}}, M), \qquad x^{\text{high}} = \mathcal{D}(z_0^{\text{high}}), \quad x^{\text{low}} = \mathcal{D}(z_0^{\text{low}}). \tag{2}$$

The two images are then blended exactly as in foveated rendering in graphics, where content is synthesised at full resolution only near the gaze and at reduced resolution elsewhere:

$$x^{\text{fov}} = M' \odot x^{\text{high}} + (1 - M') \odot \text{Up}\big(x^{\text{low}}\big), \tag{3}$$

where $\text{Up}(\cdot)$ is spatial upsampling, $M' = \text{Up}(M) \in \{0,1\}^{H\times W}$ is the mask at pixel resolution and $\odot$ is elementwise multiplication. Running this procedure with an unmodified pretrained model is called *Naïve Mixed-Resolution Denoising*. Even with positional embeddings adapted to the mixed layout, it shows "significant scale and structural inconsistencies, frequently resulting in duplicated objects or multiple entities fused together".

Foveated training and results. The paper fixes this failure with a short post-training on foveated examples, described in Sec. 3.2 together with our reproduction. With it, the method matches full-resolution generation on HPSv2.1 and CLIP score at a 43% token ratio, gives up to 2× (image) and 3.5–4× (video) speedups, and in a gaze-contingent two-alternative forced-choice study is chosen over full-resolution generation 47.3% of the time (i.e. perceptually indistinguishable) and over the naïve baseline 87.4% of the time (Fig. 2).

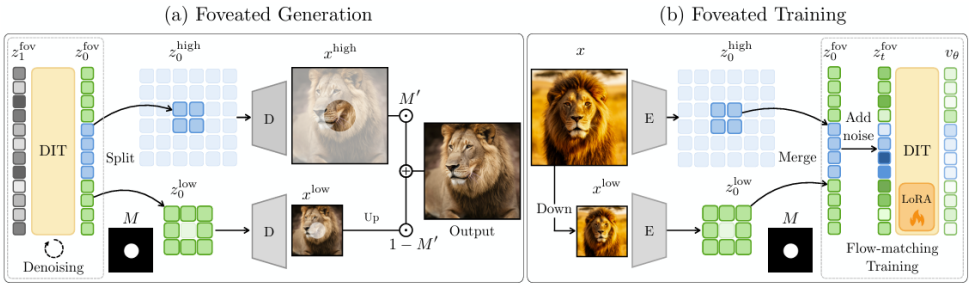

Figure 1: The Foveated Diffusion pipeline, reproduced from the original paper (Fig. 2). (a) A shortened mixed-resolution token sequence is denoised, split, decoded per resolution and blended (Eq. 3). (b) Training targets are built by encoding the image and its downsampled copy separately and merging them according to the mask.

Figure 2: Qualitative results, reproduced from the original paper (Fig. 5). The naïve mixed-resolution baseline shows scale inconsistencies and duplicated or fused objects around the fovea (white circle), while Foveated Diffusion is perceptually close to full high-resolution generation.

2 Motivation: foveation as efficient coding

In class we saw that the brain builds a neural code $z = f(x)$ of its sensory input under resource constraints. Efficient coding states this as a trade-off between fidelity and cost,

$$\min_{f,g} \; \mathbb{E}_x\big[\, d\big(x, g(f(x))\big)\big] + \lambda\, C\big(f(x)\big), \tag{4}$$

where d is the information lost by the decoder g and C the metabolic or bandwidth cost of the code. In early vision this yields centre–surround receptive fields in the retina, which respond to edges rather than uniform regions, and in V1 localized, oriented, band-pass filters well described by Gabor functions, $w(u,v) \approx \exp\big(-\frac{u'^2+\gamma^2 v'^2}{2\sigma^2}\big)\cos(2\pi f u' + \varphi)$.

Foveation is the same principle acting on *where* resources are spent. Retinal ganglion cell density, and with it acuity, drops by more than an order of magnitude from the fovea to the periphery; the eye then compensates through active sensing, moving the fovea with saccades to where the information is. Foveated Diffusion maps directly onto Eq. (4): the code z is the token sequence, the cost C is the token count (and the quadratic attention it induces), and the distortion d is *perceptual*, which is why low peripheral resolution is affordable. What the paper leaves outside the model, however, is the second half of the biological solution, deciding where the fovea should go. The mask is either supplied (eye tracking or random placement) or derived from an external deep saliency network.

This is what led us to combine the two views. Early vision tells us which parts of an image carry the information that is expensive to represent: regions with strong oriented, band-pass structure, exactly what a bank of Gabor filters responds to, while smooth regions are cheap and cancel under centre–surround filtering. A foveation policy grounded in this V1-like computation is cheap, interpretable and biologically motivated. Before pursuing that direction we first had to establish a faithful, working foveated generator, which is the subject of the next section.

3 Reproducing Foveated Diffusion

3.1 Setup and the naïve baseline

The paper fine-tunes a 4B-parameter FLUX.2 model on eight H100 GPUs. We had a single 8 GB consumer GPU, so we used PixArt-α XL-2-512 (0.6B parameters, 512×512 images, $32 \times 32 = 1024$ tokens), a pure transformer denoiser that keeps everything the method needs. It differs from FLUX in two relevant ways: it uses absolute sin–cos positional embeddings rather than rotary ones, and an older 4-channel VAE. We implemented foveated tokenization directly on PixArt's tokens. Foveal tokens are taken from the full latent and peripheral tokens from a $2\times$ coarser grid, both are packed into one sequence, and each peripheral token receives the positional embedding of the centre of the block it replaces. Decoding follows Eqs. (2)–(3). A centred fovea of radius 0.25 keeps 436 of 1024 tokens (43%), the first operating point of the paper's Table 1.

The paper's failure vs. ours. In the paper, the naïve baseline is degraded but intact: the periphery is a plausible image whose objects have the wrong scale or are duplicated across the boundary (Fig. 2, naïve columns). Ours fails differently (Fig. 3). The fovea is rendered correctly and matches the full-resolution image from the same seed, but the periphery collapses into saturated, glowing blobs of neon colour ("shining blue objects") with no recognizable structure, for every mask we tried.

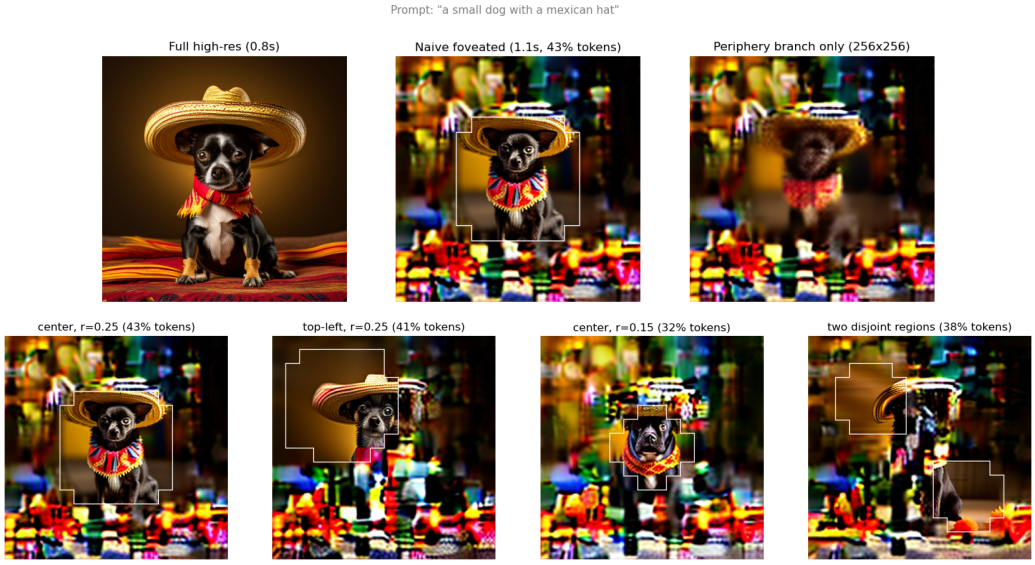

Figure 3: Our naïve mixed-resolution denoising (compare the paper's naïve columns in Fig. 2). **Top:** same prompt and seed as the full-resolution image (left); the fovea (white outline) is coherent, while the periphery degenerates into saturated colour blobs. **Bottom:** the failure persists across mask position, size and disjoint foveae.

Why ours diverges. The peripheral latents are only moderately larger than the foveal ones, and changing the guidance strength in the periphery alone does not change this, so the problem is the *structure* of the peripheral content rather than its magnitude. Two causes stand out. First, a peripheral token's positional embedding tells the model where the token is but not that it covers four times the area, so the model paints full-resolution-scale detail there, which becomes coarse, high-contrast blocks after upsampling. Second, the older 4-channel VAE handles such unfamiliar latents poorly and clips colour channels unevenly, which produces exactly the neon hues we see; the paper's modern VAE is far more forgiving.

3.2 Foveated LoRA training

Method. The failure of the naïve approach arises because the pretrained model has never encountered a foveated token sequence. Following the original paper, we address this by utilizing Low-Rank Adaptation (LoRA), which updates the layer weights using the formulation $W = W_0 + BA$. During this additional training phase,

the original pretrained weights (W_0) remain frozen to preserve the model's prior knowledge of image generation. Simultaneously, the small low-rank matrices (B and A, rank 32, comprising about 4% of the parameters) are injected into the attention and feed-forward layers across 6 specific transformer blocks, allowing them to learn the new token format. During training, the model is exposed to foveated versions of real images—sharp inside a randomly placed fovea and coarse outside it. By learning to denoise these mixed-resolution examples, the model is essentially taught to compensate for the structural artifacts caused by foveation. Since both regions originate from the same image, the network deduces that the periphery is the *same scene* at a lower detail, ensuring that objects maintain their scale and continuity across the boundary. Finally, we randomized the fovea's location between examples, with intension than the model will adapt to any arbitrary mask.

Training on a budget. We trained on 10,000 DiffusionDB images in a single pass: 10,000 steps, one step per image, each with its own random circular fovea, on our single GPU. The paper used 90,000 images and eight H100s.

Results. To the eye, the trained model fixes the naïve failure almost completely (Fig. 4, compare Fig. 2). The periphery serves as a natural, low-resolution continuation of the scene. In this example, the lion's body, mane and the horizon seamlessly continue across the fovea boundary at a consistent scale, drastically reducing the occurrence of neon blobs and duplicated structures. The peripheral branch alone is a coherent 512×512 image of the same scene. A faint halo remains where the blend crosses the smooth sky, similar to the boundary artifacts the paper reports for its own method.

A notable outcome of this implementation is the visual quality achieved despite strict resource constraints. While our scope naturally does not match the massive scale and rigorous quantitative benchmarks of the original paper, our generated foveated images remain visually compelling. We utilized a model $7\times$ smaller, approximately 11% of their training data (processed for only a single epoch), and a single consumer-grade GPU. This implies that adapting a pretrained generator to a mixed-resolution token format is a highly parameter-efficient task; the model merely learns to interpret the new layout rather than relearning image generation from scratch.

Moreover, we acknowledge several limitations in our reproduction, which manifest both in the visual outputs and our evaluation methodology. Visually, as mentioned, we observed occasional structural inconsistencies where the transition between the high-resolution fovea and the low-resolution periphery was not perfectly seamless. Additionally, the peripheral regions sometimes exhibited oversaturated, 'neon-like' color shifts. Although reducing the classifier-free guidance scale mitigated these shifts, artifacts persisted. We attribute this discrepancy from the original paper's results to the inherent bottlenecks of utilizing a weaker model, lower resolution, and limited training data. Specifically, our pre-trained 4-channel VAE struggles to decode compressed low-resolution tokens and clips their extreme latent values—a limitation effectively avoided in the original study through the use of modern, high-capacity architectures.

On the methodological front, our assessment remains entirely qualitative, lacking the formal quantitative metrics (e.g., HPSv2.1, FID, CLIP) and user studies presented in the original literature.

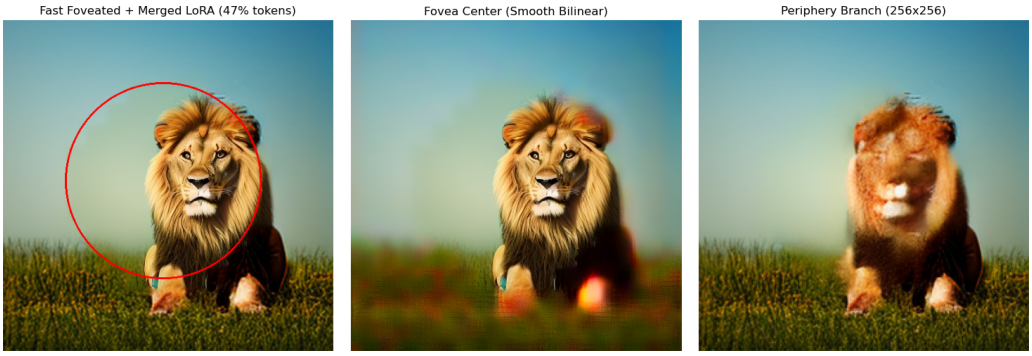

Figure 4: Our foveated generation after LoRA training (compare the paper's "Ours" columns in Fig. 2), at a 47% token ratio (red circle: fovea), next to the full-resolution and low-resolution (256×256) branches before blending. Structure and scale are consistent across the boundary; a faint blending halo remains in the sky.

4 Follow-up ideas

The paper takes the fovea as given. Following Sec. 2, we ask whether a V1-like computation can decide where it goes: a bank of Gabor filters marks the regions with strong oriented, band-pass structure, which are exactly the regions that are expensive to represent at low resolution. All three ideas below use the same *Gabor mask*: the maximum response over four orientations $(0°, 45°, 90°, 135°)$, normalised, smoothed with a Gaussian and thresholded, then reduced to the 32×32 token gri.

1. **Mid-trajectory Gabor foveation.** Start denoising at full resolution, and after a few steps decode the model's current estimate of the image, compute its Gabor mask and continue foveated with it. This needs no more training than what we have done up to this point, like a saccade that lands only after a first coarse glance, but the first steps pay the full cost.
2. **Prompt-to-mask network.** Train a small network to predict the Gabor mask of an image from its prompt alone, so the fovea is known before generation starts, analogous to top-down expectations guiding where we look.
3. **Gabor-mask LoRA.** The paper also trains *content-guided* variants, in which each training image's fovea is placed on its important content instead of at random: a saliency map predicted by the DeepGaze IIE network, or object bounding boxes detected by YOLO. Since the mask then correlates with content, the trained model learns to place salient objects inside whatever fovea it is given at inference. The cost is that a deep network must run on every training example to produce its mask, which at the paper's 90,000 images is a substantial preprocessing step of its own. We replace that network with the Gabor mask above: each training image gets its own mask from a fixed bank of four filters, a few convolutions with no learned parameters, at a negligible cost per image. We then retrain the LoRA of Sec. 3.2 with these masks instead of random circles, so the generator learns to put its edge-rich content inside the fovea, guided by an early-vision computation rather than a deep detector.

Chosen experiment: idea 3. We test whether the cheap Gabor masks are a good replacement for random foveae during training. The hypothesis is that a LoRA trained on Gabor masks produces foveated images at least as close to full-resolution generation as a LoRA trained on random circles, and far closer than the naïve baseline.

Metrics. Metrics. We evaluate our models using two primary metrics: generation time and the CLIP score. The generation time measures the end-to-end computational cost per image, validating the efficiency gains of the mixed-resolution token layout. The CLIP score assesses the semantic alignment between the generated image and the input text prompt. We selected CLIP over pixel-level reconstruction metrics because it effectively penalizes the semantic degradation caused by out-of-distribution VAE clipping (the "neon" artifacts) in the periphery, objectively quantifying whether the holistic image context survives the compression.

Setup. We conducted a robust quantitative evaluation across 30 diverse text prompts, generating 3 given seeds (which where chose at random) per prompt to obtain 90 images per configuration (total of 270 images per model). We compared four models: (1) the full high-resolution base model, (2) the naïve baseline (foveated layout without LoRA), (3) a LoRA trained on random circular masks, and (4) our LoRA trained on Gabor-filtered masks. Each foveated model was evaluated on three spatial layouts: a centered fovea ($r = 0.28$), a top-left fovea ($r = 0.28$), and two disjoint foveae ($r = 0.15$). To ensure an unbiased comparison, all generation parameters were kept strictly identical across models.

Results. Table 1 summarizes the mean and standard deviation for the token ratio, generation time, and CLIP score across all mask layouts and evaluated models. Moreover, chosen high CLIP examples for the generated images are provided in the appendix at the end of this document.

Mask	Model	Token Ratio	Seconds		CLIP Score	
			Mean	Std	Mean	Std
None (Full Resolution)	Full high-res base	1.000	2.541	0.471	32.576	2.091
Centred circle r=0.28	Naïve baseline	0.449	1.280	0.046	27.645	3.852
	Random-mask LoRA	0.449	1.255	0.014	32.814	2.200
	Gabor-mask LoRA	0.449	1.256	0.010	32.325	2.240
Top-left circle r=0.28	Naïve baseline	0.440	1.245	0.009	25.305	4.088
	Random-mask LoRA	0.440	1.251	0.010	32.501	2.208
	Gabor-mask LoRA	0.440	1.251	0.009	32.521	2.123
Two foveae r=0.15	Naïve baseline	0.359	1.081	0.008	21.487	3.966
	Random-mask LoRA	0.359	1.078	0.008	32.388	2.127
	Gabor-mask LoRA	0.359	1.081	0.008	32.235	2.315

Table 1: Quantitative evaluation across 30 prompts (3 seeds each). The full high-resolution model serves as the baseline reference. The naïve baseline suffers severe semantic degradation (low CLIP scores), whereas the LoRA models successfully restore performance to baseline levels while halving the inference time.

Insights. The quantitative results strongly support our hypothesis. First, the naïve baseline suffers a severe semantic collapse, particularly in the most complex layout (two foveae), where its CLIP score plummets to 21.487. This mathematically reflects the severe visual artifacts that destroy the peripheral context. Second, both the Random-mask and Gabor-mask LoRAs successfully restore the semantic fidelity to match the full high-resolution baseline (achieving scores between 32.2 and 32.8), despite discarding 55% to 64% of the tokens. Furthermore, the foveated models demonstrate a practical efficiency gain, cutting the end-to-end generation time by approximately 50% (from 2.54s to $\sim$ 1.25s). Finally, comparing the two training strategies reveals that the Gabor-guided LoRA performed similarly to the random-trained LoRA, with no significant quantitative advantage observed in semantic alignment. We hypothesize that this lack of distinct differentiation stems from the extreme spatial compression within the model's latent space. Because the masks are applied at a latent resolution of just 32×32, the fine-grained spatial frequencies and orientation details of the Gabor filters are heavily downsampled. At this coarse resolution, the structural nuances of the filters are largely lost, and the network likely perceives the Gabor masks as functionally similar to basic random spatial blobs, limiting their theoretical advantage as an early-vision guide.

Beyond the quantitative metrics, a qualitative analysis of the generated outputs reveals a strong correlation between the model's performance and the textural complexity of the prompt. The foveated LoRA models demonstrated significantly higher success rates when generating images dominated by low-frequency spatial details, such as broad landscapes or smooth background environments. In contrast, images requiring highly complex, high-frequency textures—such as animal fur or intricate structural patterns—were notably more challenging and exhibited more perceptible degradation in the periphery. This discrepancy suggests that the extreme spatial compression applied outside the fovea disproportionately penalizes subject matters that rely heavily on fine-grained textural fidelity, whereas broader, homogeneous scenes are inherently more robust to the loss of peripheral resolution.

Future Work. These findings suggest several promising avenues for future research. Primarily, scaling up the experiment to higher-resolution base models (e.g., models generating 1024×1024 images, which operate on a larger 128×128 latent grid) could preserve the high-frequency structural details of the Gabor filters, potentially allowing the network to fully leverage the textural priors rather than perceiving them as basic spatial blobs. Furthermore, our qualitative observations indicate that the model did not necessarily learn a strong correlation between the specific geometric shape of the mask and the structural contours of the generated object. To overcome this, future approaches could integrate the complementary strategies discussed earlier. Specifically, combining our Gabor-mask LoRA (Idea 3) with a prompt-to-mask prediction network (Idea 2) or mid-trajectory dynamic foveation (Idea 1) could provide the generator with stronger spatial expectations, explicitly forcing it to align the generated salient objects with the specific contours and layout of the foveal mask.

A Additional Generated Examples

In this appendix, we provide further qualitative examples showcasing successful image generations by our Random-mask and Gabor-mask LoRA models. These selected outputs demonstrate the models' ability to maintain high visual quality, successfully generating foveated images with coherent peripheral contexts across various diverse prompts.

Figure 5: Qualitative examples of successful generations across diverse prompts. These selected outputs highlight instances where the Random-mask LoRA high visual fidelity (passing the "eye test") alongside relatively high CLIP scores, effectively reconstructing the peripheral context without noticeable artifacts.

Figure 6: Qualitative examples of successful generations across diverse prompts. These selected outputs highlight instances where the Gabor-mask LoRA achieved high visual fidelity (passing the "eye test") alongside relatively high CLIP scores, effectively reconstructing the peripheral context without noticeable artifacts.